Facility: SALEM 1 & 2 Scenario No.: ESG-5 Op-Test No.: 13-01 NRC

Examiners: Operators:

Initial Conditions: 75% power MOL. Power was reduced 30 minutes ago due to 21 SGFP Governor problems. PZR PORV 2PR1 was declared inoperable 3 hours ago due to intermittent control circuit anomalies, and the PORV Block valve 2PR6 was shut and deenergized to comply with TSAS 3.4.5 action b.
Turnover: Maintain current power

| Event No. | Malf. No. | Event Type* | Event Description |
|---|---|---|---|
| 1 | CV0035 | C CRS/RO | Charging Master Flow Controller fails low |
| 2 | PR0017C | | Non-controlling PZR level Channel III fails (TS) |
| 3 | SG0078C | C ALL | 90 gpd SGTL (TS) |
| 4 | | R CRS/RO N PO | Downpower |
| 5 | SG0078C | M ALL | SGTR |
| 6 | AF0182B | C CRS/PO | 22 AFW pp pressure override failure |
| 7 | RC0003C | C ALL | 23 RCP trip during cooldown |
| 8 | VL0298 | C CRS/RO | PZR PORV fails shut/SGTR depress unavail →SGTR-5 |
| | | | |
| | | | CT's: #1 Isolate AFW<br>#2 C/D to, and maintain, RCS temp |
| | | | |

* (N)ormal, (R)eactivity, (I)nstrument, (C)omponent, (M)ajor

# SIMULATOR
# EXAMINATION SCENARIO GUIDE

**SCENARIO TITLE:** 13-01 NRC ESG-5

**SCENARIO NUMBER:** 13-01 NRC ESG-5

**EFFECTIVE DATE:** See Approval Dates below

**EXPECTED DURATION:** 90 minutes

**REVISION NUMBER:** 00

**PROGRAM:**

[ ] **L.O. REQUAL**

[X] **INITIAL LICENSE**

[ ] **STA**

[ ] **OTHER** ____________________

Revision Summary
New issue for 13-01 NRC Exam

| | | |
|---|---|---|
| **PREPARED BY:** | G Gauding<br>**Lead Regulatory Exam Author** | 09-14-14<br>**Date** |
| **APPROVED BY:** | Carney<br>**Operations Training Manager** | 10-23-14<br>**Date** |
| **APPROVED BY:** | MOG<br>**Facility Representative** | 10-23-14<br>**Date** |

# I. OBJECTIVES

A. Given a steam generator tube leak, take corrective action, IAW S2.OP-AB.SG-0001.

B. Given the order or indications of a steam generator tube leak (SGTL), perform actions as the nuclear control operator to RESPOND to the tube leak in accordance with the approved station procedures.

C. Given the order or indications of a steam generator tube leak (SGTL), DIRECT the response to the tube leak, in accordance with the approved station procedures.

D. Given indication of a reactor trip, DIRECT the response to the reactor trip in accordance with the approved station procedures.

E. Given the order or indications of a reactor trip perform actions as the nuclear control operator to RESPOND to the reactor trip in accordance with the approved station procedures.

F. Given indication of a safety injection DIRECT the response to the safety injection in accordance with the approved station procedures.

G. Given the order or indications of a safety injection perform actions as the nuclear control operator to RESPOND to the safety injection in accordance with the approved station procedures.

H. Given the order or indications of a steam generator tube rupture (SGTR), perform actions as the nuclear control operator to RESPOND to the tube rupture in accordance with the approved station procedures.

I. Given indication of a steam generator tube rupture (SGTR), DIRECT the response to the SGTR in accordance with the approved station procedures.

J. Given the order or indications of a steam generator tube rupture (SGTR) without pressurizer pressure control, perform actions as the nuclear control operator to RESPOND to the SGTR in accordance with the approved station procedures.

K. Given the indication of a steam generator tube rupture (SGTR) without pressurizer pressure control, DIRECT the response to the SGTR in accordance with the approved station procedures.

L. During performance of emergency operating procedures, monitor the critical safety function status trees in accordance the EOP in effect.

## II. MAJOR EVENTS

A. Charging Master Flow Controller Fails

B. Non-controlling PZR level channel fails low.

C. 90 gpd SGTL

D. SGTR

E. 22 AFW pp Pressure Overide Failure

F. 23 RCP trip during RCS cooldown

G. PORV fails shut/SGTR depressurization unavailable →SGTR-5

## III. SCENARIO SUMMARY

A. The crew will take the watch at 76% power, MOL. Power was reduced 30 minutes ago due to 21 SGFP governor problems, which have not been investigated yet. PZR PORV 2PR1 was declared inoperable 3 hours ago due to intermittent control circuit anomalies, and the PORV Block valve 2PR6 was shut and deenergized to comply with TSAS 3.4.5 action b.

B. Shortly after taking the watch, the Master Charging Flow Controller auto setpoint will fail from its current position to 0% demand over a 5 minute period, and PZR level channel III will fail low. The crew will respond IAW S2.OP-AB.CVC-0001, Loss of Charging, to ensure level channels are selected for control and alarm functions, place 23 charging pump speed controller in manual, and restore normal charging flow. If the MFC is taken to manual, the controller output will stabilize, but will be unable to be adjusted upwards. The level channel failure is silent. The CRS will identify Tech Specs.

C. After the PZR level channel Tech Spec has been identified, a small (90 gpd) SGTL will ramp in on 23 SG. The CRS will enter S2.OP-AB.SG-001, Steam Generator Tube Leak, and take actions to quantify the leak and minimize the spread of contamination in the secondary plant. The crew identifies that the SGTL meets the criteria for Action Level 3 in AB.SG, requiring the unit to be <50% power within one hour, and that Tech Spec 3.4.7.2 applies once the leak is >150 gpd.

D. Once the power reduction is underway, the affected SG tube will rupture. Operators will identify the rising leak rate, and initiate a Rx trip and Safety Injection IAW CAS when conditions warrant.

E. The crew will perform diagnostics in TRIP-1, Reactor Trip or Safety Injection. 22 AFW pump pressure override protection fails and cannot be defeated. Operators will isolate AFW to 23 SG and identify that a radioactive release from 23 SG through 23 AFW pp turbine discharge is occurring until the 23MS45, steam supply to TDAFW pump, is shut or the AFW pump is secured.

F. The crew will transition to SGTR-1, Steam Generator Tube Rupture. With the 23MS45 still open, the CRS will secure 23 AFW pump if not performed previously, leaving 24 SG as the only generator receiving AFW flow. Once secured and with 23MS45 shut, the crew will reset 23MS52, 23 AFW pump trip valve, and start 23 AFW pump.

G. During the RCS cooldown to target temperature in SGTR-1, 23 RCP will trip when the hottest CET reaches 510 degrees. This will cause a loss of normal Spray when required. 21 SI pump will not start if the crew attempts to start it, and 22 SI pump will trip if started.

H. Once the target temperature has been reached, the crew will attempt to perform the RCS depressurization with the only available PORV since 23 RCP trip caused a loss of normal spray capability, and PZR PORV 2PR1 is unavailable. 2PR2 will not open. With no SI pumps running, the CRS will transition to SGTR-5.

I. The scenario will terminate after the transition to SGTR-5 has been made.

## IV. INITIAL CONDITIONS

____ Pre-snapped IC-235

**PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)**

| *Initial* | | Description |
|---|---|---|
| ____ | 1 | RCPs (SELF CHECK) |
| ____ | 2 | RTBs (SELF CHECK) |
| ____ | 3 | MS167s (SELF CHECK) |
| ____ | 4 | 500 KV SWYD (SELF CHECK) |
| ____ | 5 | SGFP Trip (SELF CHECK) |
| ____ | 6 | 23 CV PP (SELF CHECK) |
| ____ | 7 | **2PR6 C/T** |
| ____ | 8 | Complete Attachment 2 “Simulator Ready-for-Training/Examination Checklist.” |

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

## EVENT TRIGGERS:

| Initial | ET # | Description | |
|---|---|---|---|
| | 3 | EVENT ACTION: | monp187 < 510 // HOTTEST INCORE T/C TEMP.0 |
| | | COMMAND: | |
| | | PURPOSE: | <update as needed> |

## MALFUNCTIONS:

| SELF-CHECK | Description | Delay Time | Initial Value | Ramp Time | Trigger | Severity |
|---|---|---|---|---|---|---|
| ______ 01 | SG0078C 23 STEAM GENERATOR TUBE RUPTURE | N/A | 0 | 00:05:00 | RT-1 | 0.3 |
| ______ 02 | CV0035 CHRG MASTER FLO CNTRLR FAILS H/L | N/A | 38.3 | 00:05:00 | RT-4 | 0.01 |
| ______ 03 | VL0298 2PR2 Fails to Position ( 0-100% ) | N/A | N/A | N/A | RT-3 | 0.01 |
| ______ 04 | SJ0062B 22 SAFETY INJECTION PUMP TRIP | N/A | N/A | N/A | RT-5 | |
| ______ 05 | AF0182B 22 AFP PRESS OVRD PROT FAILS | N/A | N/A | N/A | N/A | |
| ______ 06 | AN3735 AAS 735 FAILS - :21 TGA SUMP LEVEL HIGH | 00:03:00 | N/A | N/A | RT-2 | AAS POINT FAILS/OVRD TO ON |
| ______ 07 | AN3736 AAS 736 FAILS - :22 TGA SUMP LEVEL HIGH | 00:03:10 | N/A | N/A | RT-2 | AAS POINT FAILS/OVRD TO ON |
| ______ 08 | AN3737 AAS 737 FAILS - :23 TGA SUMP LEVEL HIGH | 00:03:20 | N/A | N/A | RT-2 | AAS POINT FAILS/OVRD TO ON |
| ______ 09 | AN3738 AAS 738 FAILS - :24 TGA SUMP LEVEL HIGH | 00:03:30 | N/A | N/A | RT-2 | AAS POINT FAILS/OVRD TO ON |
| ______ 10 | AN3739 AAS 739 FAILS - :25 TGA SUMP LEVEL HIGH | 00:03:42 | N/A | N/A | RT-2 | AAS POINT FAILS/OVRD TO ON |
| ______ 11 | PR0017C PZR LEVEL CH III(LT461)FAILS H/L | 00:00:01 | N/A | N/A | RT-4 | 0 |
| ______ 12 | RC0003C 23 RC PUMP ELECTRICAL TRIP | N/A | N/A | N/A | ET-3 | |

## REMOTES:

| *SELF-CHECK* | Description | Delay Time | Initial Value | Ramp Time | Trigger | Condition |
|---|---|---|---|---|---|---|
| _____ 01 | PR34D PORV STOP VALVE 2PR6 TAGGED | N/A | N/A | N/A | N/A | TAGGED |
| _____ 02 | MS06A 23MS45 23 STM GEN STM SUP-23 AFP | N/A | N/A | N/A | RT-7 | 0 |
| _____ 03 | AF01D 23 AUX FP TRIP RESET | 00:10:00 | N/A | N/A | RT-8 | RESET |

## OVERRIDES:

| *SELF-CHECK* | Description | Delay Time | Initial Value | Ramp Time | Trigger | Condition/ Severity |
|---|---|---|---|---|---|---|
| _____ 01 | AH01 F DI 21 SI PUMP STOP | 00:00:05 | N/A | N/A | ET-3 | ON |
| _____ 02 | B511 A DI 22 AUX FEED PUMP-PRESS OVERRIDE DEFEAT | N/A | N/A | N/A | N/A | OFF |

## OTHER CONDITIONS:

| | Description |
|---|---|
| _____ | 1. |

## V. SEQUENCE OF EVENTS

A. State shift job assignments.

B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).

C. Inform the crew “The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift”.

D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| **Note:** RCS temperature will be lowering ~0.1°F / minute from Xe. | | | |
| | The crew determines how much RCS dilution is required to maintain Tavg on program. | | |
| | RO initiates a dilution for RCS temperature control. | | |
| **1. Charging Master Flow Controller fails low** | | | |
| **Simulator Operator**: Insert <u>RT-4</u> on direction from Lead Evaluator.<br>**MALF: CV0035 CHRG MASTER FLO CNTRLR FAILS H/L**<br>**Severity: 0.1**<br>**Initial Value: 39.2**<br>**Ramp: 5 minutes**<br><br>**MALF: PR0017C PZR LEVEL CH III(LT461) FAILS H/I**<br>**Severity: 0** | | | |
| **Note:** There is no audible indication or letdown isolation from the PZR level Ch III failure as the high lvl/lo level alarms come off the LC460D and LC459F (channels selected for control and alarm.) | | | |
| | RO reports lowering charging flow and/or low | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | RCP seal injection flow. | | |
| | | | |
| | RO diagnoses Master Flow Controller (MFC) output lowering with PZR level on (or below) program. | | |
| | | | |
| | CRS directs RO to place Master Flow Controller in manual and restore charging flow. | | |
| | | | |
| | RO places MFC in manual, and reports MFC demand has stabilized, but cannot be raised. | | |
| | | | |
| | CRS directs RO to place 23 charging pump speed controller in manual and restore charging flow. | | |
| | | | |
| | RO reports positive control over 23 charging pump speed in manual, and raises charging flow to restore PZR level to program. | | |
| | | | |
| | CRS enters S2.OP-AB.CVC-0001, Loss of Charging, based on the reduction of charging flow. | | |
| | | | |
| | CRS directs initiation of Attachment 1 CAS. | | |
| | | | |
| | RO reports 23 charging pump in service with no indication of cavitation. | | |
| | | | |
| **2. Non-controlling PZR Level Channel III fails low.** | | | |
| | | | |
| **Note**: This is at least where the level channel failure should be identified. | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | | | |
| | RO reports PZR lvl channel III failed low. | | |
| | | | |
| **Note**: AB.CVC actions for a failed PZR level channel are written for a channel which is selected for control or alarm. If CRS bases procedure flowpath on this fact, then MFC failure is addressed at Step 105. | | | |
| | | | |
| | If not done previously, RO takes manual control of charging to maintain PZR level on program. | | |
| | | | |
| | RO reports operable channels are selected for control, alarm, and recorder. | | |
| | | | |
| | CRS determines MFC will not be returned to auto. | | |
| | | | |
| | CRS directs removal of PZR level channel III from service. | | |
| | | | |
| | CRS enters TSAS 3.3.1.1 action 6. | | |
| | | | |
| **Note:** The following steps are the actions in AB.CVC for getting to the failed MFC step105 after first reaching the step (54) for a failed PZR level channel. | | | |
| | | | |
| | RO reports VCT lvl channel has not failed. | | |
| | | | |
| | RO reports no indication of charging header | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | leak. | | |
| | | | |
| | RO reports charging MFC failed. | | |
| | | | |
| | RO reports manual control of 23 charging pump has been established, and reports PZR level can be maintained stable or rising. | | |
| | | | |
| | CRS assigns responsibility and band for manual PZR level control. | | |
| | | | |
| Proceed to next event after AB.CVC-1 has been addressed and PZR level channel failure TS has been identified at Lead Evaluators direction. | | | |
| | | | |
| **2. SGTL** | | | |
| | | | |
| **Simulator Operator**: Insert **RT-1** on direction from Lead Evaluator.<br><br>**MALF: SG0078C 23 Steam Generator Tube Rupture**<br>**Final Value: 0.3**<br>**Ramp: 15 minutes** | | | |
| | | | |
| | RO announces OHA A-6 RMS HI RAD OR TRBL as unexpected. | | |
| | | | |
| | RO reports CRT shows 2R53C in alarm. | | |
| | | | |
| | RO reports 2R53C reading and slowly rising. | | |
| | | | |
| | CRS contacts Radiation Protection to perform | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | SC.RP-TI.RM-0607(Q), Primary To Secondary Leak Rate Response IAW ARP. | | |
| | | | |
| | CRS enters S2.OP-AB.SG-001 Steam Generator Tube Leak directly or enters S2.OP-AB.RAD-001, Abnormal Radiation | | |
| | | | |
| | CRS directs performance of S2.OP-AB.SG-001 CAS. | | |
| | | | |
| Go to page 14 when Action Level 3 is identified. | | | |
| | | | |
| | | | |
| | CRS dispatches an operator to deenergize TGA sumps. | | |
| | | | |
| **Simulator Operator**: Insert **RT-2** when directed to deenergize TGA sumps. This RT includes a 3 minute delay before opening sump breakers. Report to control room when last of the 5 sump breakers has been deenergized.<br>**MALFS:**<br>**AN3735 21 TGA Sump Level Hi**<br>**AN3736 22 TGA Sump Level Hi**<br>**AN3737 23 TGA Sump Level Hi**<br>**AN3738 24 TGA Sump Level Hi**<br>**AN3739 25 TGA Sump Level Hi**<br>**Delays: 3:00, 3:10, 3:25, 3:37, 3:57** | | | |
| | | | |
| | RO reports PZR level is stable. IF RO reports PZR level is lowering, RO will swap to a | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | centrifugal charging pump. | | |
| | | | |
| | RO reports unit is in Mode 1. | | |
| | | | |
| | Crew identifies 2R53C, 2R19C and 2R15 rising. | | |
| | | | |
| | PO reports affected SG is 23. | | |
| | | | |
| | PO raises 23MS10 setpoint to 1045 psig. | | |
| | | | |
| | PO closes or checks closed 23GB4, 23MS7, and 23MS18. | | |
| | | | |
| | CRS dispatches an operator to shut 23MS45, and enters TSAS 3.7.1.3 for 23 AFW pump when required. | | |
| | | | |
| **Simulator Operator:** Do **NOT** shut 23MS45 until directed later in scenario. | | | |
| | | | |
| | CRS dispatches an operator to re-align SGBD and MS sampling to Waste System. | | |
| | | | |
| **Simulator Operator:** 10 minutes after chemistry is first contacted, report that 23 SG has developed a 300 gpd leak, from 0 primary to secondary previously. | | | |
| | | | |
| CAS Action Level 3 required actions start here. | | | |
| | | | |
| | CRS determines that the CAS indicates that | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | Action Level 3 is present IAW Step 6.2.A based on:<br>- Leak rate is $\geq$ 75 gpd AND<br>- The rate of change of the leakrate is $\geq$ 30 gpd/hr. | | |
| | | | |
| | CRS determines a power reduction to $\leq$50 % must be performed within 1 hour. | | |
| | | | |
| | CRS enters TSAS 3.4.7.2.c | | |
| | | | |
| **Note**: If time does not permit Tech Spec identification, post scenario follow up question will be required. | | | |
| | | | |
| **4. Downpower** | | | |
| | | | |
| | RO calculates boron addition required for power reduction to 50%. | | |
| | | | |
| | CRS orders a power reduction at a rate which will ensure power is <50% within one hour. | | |
| | | | |
| | CRS enters S2.OP-AB.LOAD-0001, Rapid Load Reduction to perform the power reduction. | | |
| | | | |
| | PO initiates a turbine load reduction at rate specified by CRS | | |
| | | | |
| | RO initiates RCS boration at rate directed by CRS. | | |
| | | | |
| | RO announces expected and actual rod movement when it occurs. | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | | | |
| Proceed to next event on direction from Lead evaluator. | | | |
| | | | |
| **5. 23 SGTR** | | | |
| | | | |
| **Simulator Operator: MODIFY MALF SG0078C to 650** with no ramp or delay after power reduction has commenced or at direction of Lead Eval, but AFTER SG tube leak exceeds 150 gpd. (TS threshold) | | | |
| | | | |
| | RO reports indications of worsening tube leak on 23 SG. | | |
| | | | |
| | CRS determines CAS actions for rising SG NR level IAW CAS 1.0 are true. | | |
| | | | |
| | CRS directs the RO to trip the Rx, confirm the trip, and initiate a Safety Injection. | | |
| | | | |
| | RO trips the Rx, confirms the trip, and initiates a Safety Injection. | | |
| | | | |
| | RO performs immediate actions of TRIP-1:<br><br>Verifies Rx tripped.<br>Verifies Rx trip is confirmed.<br>Backs up Main Turbine trip.<br>Verifies off site power available to at least one vital buss.<br>Verifies SI initiation. | | |
| | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | CRS reads immediate action steps to RO who confirms their performance. | | |
| | | | |
| | Crew commences monitoring TRIP-1 CAS. | | |
| | | | |
| | RO reports SEC loading is not complete for all vital busses, but all available equipment started. | | |
| | | | |
| **6. 22 AFW pump Pressure Override circuit malfunction** | | | |
| | | | |
| | PO reports all AFW pumps are running, but 22 AFW pump is not supplying flow even though its discharge pressure is high enough. | | |
| | | | |
| | PO requests, and receives, permission to depress Pressure Override Defeat for 22 AFW pump, which has no effect on AFW flow. | | |
| | | | |
| | PO reports 23 AFW pump is running, and that an unmonitored release is occurring from the TDAFW pump steam discharge with 23MS45 not shut yet. | | |
| | | | |
| AFW flow control is complicated by the fact that a SGTR is present on 23 SG, but the 23MS45 has not been shut yet. Additionally, 22 AFW pump is not supplying flow to 21 and 22 SGs because its Pressure Override circuit has failed. Shortly after the Rx is tripped, SG level will recover in at least one SG so that AFW flow can | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| be lowered < 22E4. TRIP-1 does not direct tripping 23 AFW pump unless 2 MDAFW pps are running, which in this case is not true, since a running pump would be supplying flow. Detailed observation of the crew during AFW flow decisions should be made by the evaluators. | | | |
| | | | |
| | | | |
| **CT# 1: (CT-18) Isolate AFW to the ruptured SG within 10 minutes of entry into TRIP-1 and subsequently close 23MS167, 23MS18, 23MS7 and 23GB4 before a transition to SGTR-3 is required.**<br><br>**SAT_____ UNSAT_____** | | | |
| | | | |
| | PO requests to throttle AFW flow, and isolates AFW flow to 23 SG by closing 23AF11 and 23AF21. | | |
| | | | |
| | CRS directs PO to maintain total AFW flow >22E4 lbm/hr while throttling AFW flow. | | |
| | | | |
| | RO reports normal containment pressure. | | |
| | | | |
| | CRS determines no MSLI is required. | | |
| | | | |
| | CRS directs SM to refer to the ECG. | | |
| | | | |
| | PO reports all 4KV vital busses energized from off site power. | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | | | |
| | RO reports control room ventilation in accident pressurized mode. | | |
| | | | |
| | RO reports 2 CCW pumps are running. | | |
| | | | |
| | RO reports ECCS is injecting as expected for current RCS pressure. | | |
| | | | |
| | PO reports AFW flow and SG NR level status. | | |
| | | | |
| | RO reports all RCPs are in operation. | | |
| | | | |
| | RO reports MSLI is not required. | | |
| | | | |
| | RO reports RTBs are open, PORVs shut, PORV block valve 2PR7 open, 2PR6 shut and C/T, and PZR spray valves operating as expected for current RCS pressure. | | |
| | | | |
| | RO reports RCS pressure > 1350 psig. | | |
| | | | |
| | RO maintains seal injection flow to all RCPs. | | |
| | | | |
| | PO reports no faulted SGs. | | |
| | | | |
| | PO reports 23 SG is ruptured based on NR level. | | |
| | | | |
| | CRS transitions to SGTR-1. | | |
| | | | |
| | RO maintains seal injection flow to all RCPs. | | |
| | | | |
| | PO reports 23 SG is ruptured, and 23MS10 is | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | set at 1045 psig. | | |
| | | | |
| | PO reports 23MS10 operating as expected for current pressure. | | |
| | PO ensures 23MS167, 23MS7, 23MS18, and 23GB4 are shut. | | |
| | | | |
| | PO reports 23 SG is ruptured. | | |
| | | | |
| | PO reports 23 AFW pump is NOT only source of AFW. | | |
| | | | |
| | PO lowers 23 AFW pump speed to minimum and trips 23 AFW pump. | | |
| | | | |
| | PO stops 23 AFW pump. | | |
| | | | |
| **Simulator Operator:** When 23 AFW pump is stopped, and an operator has previously been dispatched, insert **RT-7** to close 23MS45 and call control room to inform them it is shut.<br><br>**REMOTE: MS06A 23MS45 23 STM GEN STM SUP-23 AFP** | | | |
| | | | |
| | CRS sends an operator to reset 2MS52 when all SG NR levels are > 15%. | | |
| | | | |
| **Simulator Operator**: Insert **RT-8** to reset 2MS52. RT includes a 10 minute delay prior to resetting MS52. | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| **REMOTE: AF01D 23 AUX FP TRIP RESET Final Value: RESET** | | | |
| | | | |
| | CRS checks on status of MS sampling valves which were directed to be shut in AB.SG. | | |
| | | | |
| When contacted by crew report 2SS333 is shut. | | | |
| | | | |
| | PO reports 23 SG is isolated from intact SGs, 23 SG NR level is >9%, and feed flow is isolated to 23 SG. | | |
| | | | |
| | RO reports power is C/T to 2PR6, and CRS either leaves it C/T, or orders power restored. | | |
| | | | |
| **Note**: Do **NOT** restore power to 2PR6 during scenario. If asked later in scenario, state you are having problems with WCM getting the release authorized. | | | |
| | | | |
| | RO reports 2PR2 is operating correctly in response to PZR pressure. | | |
| | | | |
| | PO reports no faulted SGs. | | |
| | | | |
| | RO resets SI and Phase A isolations, and opens CA330's. | | |
| | | | |
| | PO resets all SECs. | | |
| | | | |
| | RO stops both RHR pumps. | | |
| | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | PO reports 23 SG is isolated and >375 psig. | | |
| | | | |
| | CRS dispatches an operator to shift Gland Sealing steam to Unit 1. | | |
| | | | |
| | CRS determines target temperature for RCS cooldown is 503 degrees. | | |
| | | | |
| | PO commences RCS cooldown by using Steam Dumps in MS Pressure Control-Manual at 25% demand. | | |
| | | | |
| | PO Bypasses Tavg when Tavg reaches 543. | | |
| | | | |
| **7. Loss of Off-site power** | | | |
| | | | |
| **Simulator Operator**: Ensure **ET-3** is TRUE when the hottest CET reaches 510 degrees. This inserts the 23 RCP trip..<br><br>**MALF: RC0003C 23 RC PUMP ELECTRICAL TRIP**<br><br>**I/O AH01 OVDI 21 SI Pump STOP** | | | |
| | | | |
| | RO reports 23 RCP trip. | | |
| | | | |
| | RO shuts charging pump mini flows when RCS pressure lowers to 1500 psig. | | |
| | | | |
| | PO stabilizes hottest CET temperature less than 503 degrees. | | |
| | | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| **CT #2 (CT-19):** Establish/maintain an RCS temperature so that transition from SGTR-1 is not required either because minimum required subcooling cannot be maintained, or because RCS low temperature causes a RED or PURPLE challenge to the subcriticality and/or the integrity CSF.<br><br>**SAT_____ UNSAT_____** | | | |
| | | | |
| | PO reports 23 SG pressure is stable or rising or at least 250 psig above intact SG pressure. | | |
| | | | |
| | RO reports normal spray is not available due to 23 RCP trip. | | |
| | | | |
| **Simulator Operator**: Insert **RT-3** when Step 18 PORV status is read PRIOR to crew attempting to open 2PR2.<br>**MALF: VL0298 2PR2 fails to position (0-100%)**<br>**Final Value: 0.01** | | | |
| | | | |
| | RO reports 2PR2 PZR PORV is available. | | |
| | | | |
| **8. PZR PORV fails shut** | | | |
| | | | |
| | RO reports 2PR2 will not open | | |
| | | | |
| | CRS returns to step 18 and answers NO to if a | | |

| Evaluator/Instructor Activity | Expected Plant/Student Response | SBT LOG | Comment |
|---|---|---|---|
| | PORV is available. | | |
| | | | |
| **Simulator Operator**: IF CRS did not start SI pumps subsequent to the Blackout, and starts them here, THEN insert **RT-5** after 22 SI pump has been started.<br>**MALF: SJ0062B 22 Safety Injection Pump trip.**<br>**Note:** 21 SI pump will not start. | | | |
| | | | |
| | RO reports no SI pumps are running. | | |
| | | | |
| | RO reports no SI pumps running if start attempt is made. | | |
| | | | |
| | CRS transitions to SGTR-5. | | |
| | | | |
| Terminate scenario after transition to SGTR-5 in announced. | | | |
| | | | |
| | | | |
| | | | |
| | | | |

## VI. SCENARIO REFERENCES

A. Alarm Response Procedures (Various)
B. Technical Specifications
C. Emergency Plan (ECG)
D. OP-AA-101-111-1003, Use of Procedures
E. S2.OP-IO.ZZ-0004, Power Operation
F. S2.OP-AB.CVC-0001, Loss of Charging
G. S2.OP-AB.SG-0001, Steam Generator Tube Leak
H. S2.OP-AB.LOAD-0001 Rapid Load Reduction
I. 2-EOP-TRIP-1, Reactor Trip or Safety Injection
J. 2-EOP-SGTR-1, Steam Generator Tube Rupture
K. 2-EOP-SGTR-5, SGTR without Pressurizer Pressure Control

## ATTACHMENT 1
## UNIT TWO PLANT STATUS
## TODAY

MODE: 1 POWER: 76 RCS BORON: 885 MWe 920

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

Core burnup is 5,000 EFPH
Control Bank D rods are at 148 steps
Power was reduced to 76% 30 minutes ago due to slight oscillations on 21 SGFP governor.
Xe is building in at 60 pcm / hr.

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

3.4.5. Action b- 2PR1 expires 69 hours from now

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

ABNORMAL PLANT CONFIGURATIONS:

2PR6 shut and power C/T.

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.
No penalty minutes in the last 24 hrs.

PRIMARY:

PZR PORV 2PR1 was declared inoperable 3 hours ago due to intermittent control circuit anomalies.

SECONDARY:

21 SGFP remains in service. Governor problem has not been identified.
Heating steam is aligned to Unit 1.

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

None

## ATTACHMENT 2

## SIMULATOR READY FOR TRAINING CHECKLIST

____ 1. Verify simulator is in "TRAIN" Load

____ 2. Simulator is in RUN

____ 3. Overhead Annunciator Horns ON

____ 4. All required computer terminals in operation

____ 5. Simulator clocks synchronized

____ 6. All tagged equipment properly secured and documented

____ 7. TSAS Status Board up-to-date

____ 8. Shift manning sheet available

____ 9. Procedures in progress open and signed-off to proper step

____ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced

____ 11. Required chart recorders advanced and ON (proper paper installed)

____ 12. All printers have adequate paper AND functional ribbon

13. Required procedures clean

____ 14. Multiple color procedure pens available

____ 15. Required keys available

____ 16. Simulator cleared of unauthorized material/personnel

____ 17. All charts advanced to clean traces and chart recorders are on.

____ 18. Rod step counters correct (channel check) and reset as necessary

____ 19. Exam security set for simulator

____ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter

____ With Baseline Data filled out

____ 21. Shift logs available if required

____ 22. Recording Media available (if applicable)

____ 23. Ensure ECG classification is correct if applicable.

____ 24. Reference verification performed with required documents available

____ 25. Verify phones disconnected from plant after drill.

____ 26. Verify EGC paperwork is marked "Training Use Only" and is current revision.

____ 27. Ensure sufficient copies of ECG paperwork are available.

## ATTACHMENT 3

## CRITICAL TASK METHODOLOGY

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.
The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...

- degradation of any barrier to fission product release
- degraded emergency core cooling system (ECCS) or emergency power capacity
- a violation of a safety limit
- a violation of the facility license condition
- incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
- a significant reduction of safety margin beyond that irreparably introduced by the scenario

II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...

- effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
- recognize a failure or an incorrect automatic actuation of an ESF system or component.
- take one or more actions that would prevent a challenge to plant safety.
- prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation.

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

**SCENARIO IDENTIFIER:** 13-01 NRC-ESG-5 **REVIEWER: P Williams**

| Initials | | Qualitative Attributes |
|---|---|---|
| PW | 1. | The scenario has clearly stated objectives in the scenario. |
| PW | 2. | The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events. |
| PW | 3. | The scenario consists mostly of related events. |
| PW | 4. | Each event description consists of:<br>• the point in the scenario when it is to be initiated<br>• the malfunction(s) that are entered to initiate the event<br>• the symptoms/cues that will be visible to the crew<br>• the expected operator actions (by shift position)<br>• the event termination point |
| PW | 5. | No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event. |
| PW | 6. | The events are valid with regard to physics and thermodynamics. |
| PW | 7. | Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives. |
| PW | 8. | The simulator modeling is not altered. |
| PW | 9. | All crew competencies can be evaluated. |
| PW | 10. | The scenario has been validated. |
| PW | 11. | If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario. |
| PW | 12. | ESG-PSA Evaluation Form is completed for the scenario at the applicable facility. |

## ATTACHMENT 4

## SIMULATOR SCENARIO REVIEW CHECKLIST

Note: The quantitative attribute target ranges that are specified on the form are not absolute limitations; some scenarios may be an excellent evaluation tool, but may not fit within the ranges. A scenario that does not fit into these ranges shall be evaluated to ensure that the level of difficulty is appropriate. (ES-301 Section D.5.d)

| Initial | | Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d) |
|---|---|---|
| GG | 7 | Total malfunctions inserted: 5-8 |
| GG | 3 | Malfunctions that occur after EOP entry: 1-2 |
| GG | 2 | Abnormal Events: 2-4 |
| GG | 1 | Major Transients: 1-2 |
| GG | 2 | EOPs entered/requiring substantive actions: 1-2 |
| GG | 1 | EOP contingencies requiring substantive actions: 0-2 |
| GG | 2 | Crew Critical Tasks: 2-3 |

**COMMENTS:**

13-01 NRC ESG-5

**CT# 1: (CT-18) Isolate AFW to the ruptured SG within 10 minutes of entry into TRIP-1 and subsequently close 23MS167, 23MS18, 23MS7 and 23GB4 before a transition to SGTR-3 is required.**

**Basis:** Failure to isolate the ruptured SG causes a loss of differential pressure between the ruptured SG and the intact SGs. The fact that the crew allows the differential pressure to dissipate and, as a result, are then forced to transition to a contingency ERG constitutes an incorrect performance that "necessitates the crew taking compensating action that would complicate the event mitigation strategy...."

**CT#2(CT-19) Establish/maintain an RCS temperature so that transition from E-3 does not occur because the RCS temperature is in either of the following conditions:**

- **Too high to maintain minimum required subcooling**

**OR**

- **Below [the RCS temperature that causes an RED path or a PURPLE challenge to the subcriticality and/or the integrity CSF.**

**Basis:** Failure to establish and maintain the correct RCS temperature during a SGTR leads to a transition from E-3 to a contingency ERG. This failure constitutes an incorrect performance that "necessitates the crew taking compensating action that would complicate the event mitigation strategy...."

**EVENTS LEADING TO CORE DAMAGE**

| Y/N | Event | Y/N | Event |
|---|---|---|---|
| N | TRANSIENTS with PCS Unavailable | N | Loss of Service Water |
| Y | Steam Generator Tube Rupture | N | Loss of CCW |
| N | Loss of Offsite Power | N | Loss of Control Air |
| N | Loss of Switchgear and Pen Area Ventilation | N | Station Black Out |
| N | LOCA | | |

**COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY**

| Y/N | COMPONENT, SYSTEM, OR TRAIN | Y/N | COMPONENT, SYSTEM, OR TRAIN |
|---|---|---|---|
| N | Containment Sump Strainers | N | Gas Turbine |
| N | SSWS Valves to Turbine Generator Area | N | Any Diesel Generator |
| N | RHR Suction Line valves from Hot Leg | Y | Auxiliary Feed Pump |
| N | CVCS Letdown line Control and Isolation Valves | N | SBO Air Compressor |

**OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE**

| Y/N | OPERATOR ACTION |
|---|---|
| N | Restore AC power during SBO |
| N | Connect to gas turbine |
| N | Trip Reactor and RCPs after loss of component cooling system |
| N | Re-align RHR system for re-circulation |
| N | Un-isolate the available CCW Heat Exchanger |
| N | Isolate the CVCS letdown path and transfer charging suction to RWST |
| Y | Cooldown the RCS and depressurize the system |
| Y | Isolate the affected Steam Generator that has the tube rupture(s) |
| N | Early depressurize the RCS |
| N | Initiate feed and bleed |

Complete this evaluation form for each ESG.